



SHIVNETRA 47

SHIV-NETRA47 AIR DEFENCE SYSTEM PRIVATE LIMITED  
AUTONOMOUS AIR SYSTEMS • SWARM INTELLIGENCE • TACTICAL LOGISTICS

MAKE IN INDIA • ATMANIRBHAR BHARAT

Engineering Collaboration: BCN In-House Technology Team

## PART I: TACTICAL SWARM &amp; OVERWATCH ISTAR • SECTION 01

SOVEREIGN BORDER DOCTRINE  
& MULTI-TIER AERIAL ARCHITECTURE

An indigenous operational framework engineered for the unforgiving geography of India's forward northern and western frontiers—unifying expendable frontline swarm intelligence, persistent overwatch relays, and runway-independent cargo VTOL resupply under contested electronic warfare conditions.

## THE OPERATIONAL CHALLENGE

## CONTESTED MOUNTAIN FRONTIER

## HIGH-ALTITUDE HIMALAYAN FRICTION

- ✗ **Severe Himalayan Environment:** High-altitude terrain (8,000–16,000 ft AMSL), thin air reducing rotor efficiency, gale-force wind shears, and sub-zero temperatures down to -25°C.
- ✗ **Contested Electronic Warfare:** Adversary wideband RF jamming, satellite navigation (GNSS) spoofing, and severe communication attenuation through deep mountain valleys.
- ✗ **Runway Infrastructure Deficit:** Isolated frontline outposts, treacherous ridgelines, and glacial passes lack paved airstrips, making conventional fixed-wing UAVs entirely unviable.
- ✗ **High-Risk Manned Sorties:** Reliance on piloted helicopters incurs extreme financial expense, vulnerable flight paths to MANPADS threats, and weather grounding risks.

## SOVEREIGN AIR ARCHITECTURE

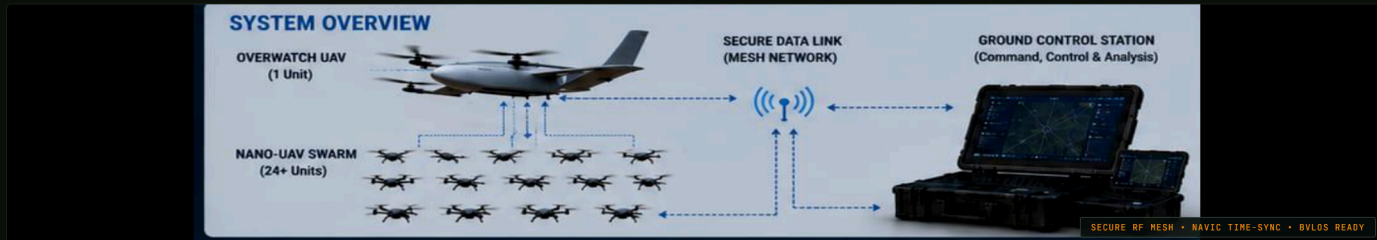
## MULTI-TIER INTEGRATION

## AUTONOMOUS DECENTRALIZED TEAMING

- ✓ **Zero Runway Dependence:** 100% vertical takeoff and landing (VTOL) enables autonomous launch and recovery from small rocky outpost pads, trucks, or forward bases.
- ✓ **Self-Healing Swarm Mesh:** Decentralized peer-to-peer RF data routing ensures continuous telemetry and video streaming even if multiple nodes are jammed or destroyed.
- ✓ **Anti-Jam NAVIC Navigation:** Autonomous flight control coupled to indigenous NAVIC satellite synchronization with fallback optical visual-inertial odometry (VIO).
- ✓ **Cross-Functional Payload Architecture:** Swappable sensor bays and cargo modules allowing tactical transitions between ISR overwatch and urgent trauma resupply.

## SYNCHRONIZED MULTI-TIER BATTLEFIELD SYSTEM ARCHITECTURE

## TACTICAL SCHEMATIC • DISTRIBUTED C2 NETWORK



## TIER 01 • FRONTLINE SCOUT

## MICRO SWARM

## NANO-SCOUT SWARM NODE

Ultra-compact, low-acoustic micro UAVs deployed in 6 to 24+ unit swarms for immediate trench clearance, perimeter overwatch, and frontline target handoff without exposing operators.

## AUV / WEIGHT

350-450 g

## LOITER TIME

25-35 Min

## OPERATING BAND

Mesh ISM / UHF

## PAYLOAD

Micro EO / IR

## TIER 02 • ENDURANCE RELAY

## TACTICAL OVERWATCH

## OVERWATCH ISTAR RELAY

Medium-range, high-altitude tactical air vehicle acting as an airborne communication relay and persistent surveillance platform equipped with gyro-stabilized optical/thermal sensors.

## PLATFORM MTOW

~20 kg Class

## SERVICE CEILING

16,000 ft AMSL

## MISSION RADIUS

50-80 km Link

## SENSOR GIMBAL

EO/IR + LRF

## TIER 03 • TACTICAL LOGISTICS

## CARGO VTOL

## HEAVY HYBRID LOGISTICS VTOL

Runway-free hybrid VTOL delivering vital blood plasma, emergency munitions, and critical spares to high-altitude outposts with precision winch drop and parachute recovery.

## PAYLOAD CAPACITY

5-10 to 200 kg

## TRANSIT RANGE

100+ km Transit

## CRUISE SPEED

120-180 km/h

## POWER SYSTEM

Hybrid electric

16,000 ft

## HIGH-ALTITUDE OPERATIONAL CEILING

Aerodynamically tailored high-aspect wings and optimized rotor profiles for thin-air mountain passes.

-25°C to +55°C

## EXTREME THERMAL ENVELOPE

Sub-zero battery thermal management coupled with desert dust and monsoon water-ingress protection.

45 km/h

## MOUNTAIN WIND SHEAR RESISTANCE

Autonomous dynamic rotor torque balancing resisting severe mountain valley updrafts and gusts.

100% Runway-Free

## TACTICAL VTOL DEPLOYMENT

Independent vertical ascent and landing from isolated border posts, warship decks, or light vehicles.